A-A-54473
19 August 1991

COMMERCIAL ITEM DESCRIPTION

CENTRIFUGE, LABORATORY (Hematology)

The General Services Administration has authorized the use of this Commercial Item Description in preference to Military Specification MIL-C-36416C which is cancelled.

This Commercial Item Description covers a portable two speed centrifuge for use in blood cell washing procedures.

Salient Characteristics:

Shall be a small compact portable two speed centrifuge that is specifically designed for use in blood grouping, typing, crossmatching, the Coombs test, and other blood cell washing procedures.

Shall be designed to operate with removable and interchangeable six (6) and twelve (12) place heads that accommodate 10 X 75 millimeter (mm), 12 X 75 mm, 12 X 75 mm (Kahn) size test tubes, and other size test tubes not exceeding 12.7 X 75 mm in size. The centrifuge shall be supplied with a twelve (12) place head meeting requirements contained in this specification.

Shall comply with the applicable requirements contained in Underwriters' Laboratories UL 1262 (Standard for Laboratory Equipment).

Shall be suitable for operation on 50 and 60 Hz AC supply lines. The applicable line voltage shall be as indicated in the contract or purchase order (see "Ordering data"). A reduction (e.g. 5/6 reduction) in motor speed and timer performance on 50 Hz supply lines from that on 60 Hz shall be acceptable.

AMSC N/A FSC 6640

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Shall be designed for a two (2) speed operation that is controlled by a selector switch. The centrifuge motor shall be controlled by a timer, and a pressure-type safety switch that is in the cover latch assembly. The centrifuge motor shall only operate when the centrifuge cover is properly closed and latched, and the timer is not in the zero (0) position. The two speeds for driving a fully loaded 12 place head at 60 Hz shall be 3,300 minimum revolutions per minute (rpm) for the high speed operating mode, and 2400 +/- 100 rpm for the low speed operating mode. The centrifuge with the motor running and a balanced load shall operate without any tendency to creep.

Shall consist of a base housing, a guard bowl with a cover and latch assembly, a removable twelve (12) place head, a motor assembly, motor operation controls and a power line cord with plug termination. The motor operation controls shall include a timer, speed selector switch, cover safety switch, electric brake circuitry with selector switch, and a speed governor. The base housing shall be made of cast iron and shall be fitted with cushioning rubber feet on the underside. The manually operated switches shall be mounted in cutouts in the base housing. The centrifuge shall have overall dimensions, with cover in the closed position, that does not exceed 12 inches in height. The base dimensions shall project within a horizontal surface area that is not greater than 10 X 10 inches. The weight of a total assembled centrifuge shall not exceed 25 pounds.

Shall be supplied with a detachable twelve (12) place rotating head that is fabricated of lightweight impact resistant plastic. The tube slots shall be numbered from 1 to 12, and shall include an indexing reference so that the head after removal from the centrifuge can be returned to its original orientation on a matching board or holder (a matching board or holder is not required to be supplied with the centrifuge). The head shall be attached to a spindle located on the shaft of the centrifuge motor. The head shall also be readily removable from the spindle holder. The head shall be designed to hold the test tubes in vertical or slightly tilted positions when the centrifuge is not operating (i.e. stationary position). The centrifuging operation shall result in the test tube bottoms swinging from the stationary position to an approximate 45 degrees angle and rotating in a circular orbit having a minimum 2-3/4 inch radius. The head shall be suitable for use outside of the centrifuge as a basket or transport rack for the test tubes. The head shall also allow all the test tubes to be held vertically in place by an elastic band, so that when used a

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transportation rack the banded test tubes can be emptied simultaneously by inverting the head (i.e. pouring off supernatant liquid after centrifuging) The head shall include date coded marking (e.g. date of manufacture) to facilitate monitoring its usage with time. The manufacturer's or contractor's service data shall include a recommendation that indicates that the head should be replaced after five (5) years of usage.

Shall be supplied with an induction type motor that has a minimum rating of 1.6 amperes. The motor shall include a blower to create an air stream through the motor housing for cooling purposes.

Shall be supplied with a guard bowl that is fabricated of aluminum alloy or corrosion-resisting steel. The guard bowl shall be mounted above the base for use as a protective housing for the head. The guard bowl shall be cylindrical with a nominal eight (8) inch diameter. The guard bowl shall include a top mounted hinged cover that is transparent and fabricated of methyl methacrylate or polycarbonate. The cover shall include a suitable cutout (hole or holes) that allows for proper cooling of the spindle (holding the rotating head) during centrifuging. The guard bowl and cover shall include a latch assembly with manual release. In closure the latch assembly shall secure the closed cover and prevent it from opening when the latch has not been manually released. The closed cover shall also rest on a gasket that shall cover the exposed edge of the guard bowl. The latch assembly shall include a pressure-type safety switch that shall control the input electric power to the centrifuge motor, so that the centrifuge will only operate when the cover is both closed and properly latched.

Shall be supplied with the following electrical controls and components:

An adjustable three (3) minute (minimum range) rotary electric timer that includes a hold position for continuous operation, five (5) second graduations from 0 to 1 minutes, 15 second graduations from 1 to 3 minutes, and an "off" position which results in no electric power to the motor.

A two (2) speed selector switch for selecting the desired centrifuge motor speed mode.

An internal mounted centrifugal switch governor that controls the centrifuge motor speed when the selector switch is positioned for the low speed mode.

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A pressure-type safety switch that is included in the cover latch assembly.

An electric brake circuit with manual actuating switch for shortening the deceleration of the rotating head after the timer reaches the "off" position and shuts off electric power to the motor.

A minimum five (5) foot exposed length three-wire power cord that terminates with a three-blade male grounding plug that is suitable for use with receptacles shown in ANSI Standard C73.11 or shown in Configuration No. 5-15 of NEMA Standard WD6-88.

Shall be supplied with a manual(s) which contains complete instructions for the operation and maintenance of the centrifuge.

Workmanship. The centrifuge shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA) One centrifuge, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Inspections and testing. Inspections and testing shall be conducted in accordance with commercial practice. Sampling for the inspections and testing shall be in accordance with commercial practice. In addition, each centrifuge shall be tested to demonstrate its capability for proper operation.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Fire and casualty hazard certification. The contractor shall submit proof prior to delivery that the product offered conforms to the applicable requirements of UL 1262. Proof shall consist of one of the following:

UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be

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supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts lists, all appropriately identified.

Compliance with the requirements of UL 1262 shall not excuse compliance with other requirements of this specification and the contract or purchase order.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Other publications. The following documents form a part of this specification to the extent that is specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11 Plugs and Receptacles - General Purpose 125 Volts, 15 Amperes, 2 Pole, 3 Wire Grounding Type, Dimensions of

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

WD 6-88 Wiring Devices - Dimensional Requirements (Revision and Redesignation of ANSI C73-73)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW Suite 300, Washington, DC 20037)

UNDERWRITERS' LABORATORIES, INC.

UL 1262 Standard for Laboratory Equipment

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11746, 207 East Ohio Street, Chicago, IL 60611, and 1655 Scott Boulevard, Santa Clara, CA 95050).

Ordering data (Required line voltage, intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.)

NOTE: The following National Stock Numbers are covered by this document

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6640-00-145-1180	CENTRIFUGE, LABORATORY, 120 Volt, 60 Hz, AC
6640-01-204-8691	CENTRIFUGE LABORATORY, 120/230 Volt, 50/60 Hz, AC
6640-01-314-6713	CENTRIFUGE, LABORATORY, 120/230 Volt, 50/60 Hz, AC

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY.

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project Number: 6640-1724
Location: ENABLE/CENTRCID/S31

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54473	19 August 1991

3. DOCUMENT TITLE

CENTRIFUGE, LABORATORY (Hematology)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, MI)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center	(215)737-2870	444-2870
ADDRESS (Include Zip Code) ATTN: DPSC-MSE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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